

Michael Lindsey

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Research Summary

Applied mathematician working across scientific computing, physical simulation, and machine learning. I develop algorithms for stochastic simulation, structured compression, optimization, and scientific machine learning, with applications in quantum chemistry, condensed matter physics, chemical kinetics, RNA design, PDEs, uncertainty quantification, neural network training, and generative modeling.

Research Vision

Computational science is entering a period in which large models and agentic systems will increasingly mediate method development, implementation, analysis, and scientific decision-making.

My work has focused on core ingredients of the computational science stack:

- stochastic dynamics, rare-event sampling, and reduced models for chemical systems
- scalable numerical methods for electronic structure and scientific simulation
- high-dimensional inference, sampling, and generative modeling

I want to help build AI systems that interface reliably with real scientific workflows.

Selected Technical Contributions

These projects are unified by an interest in turning hard scientific modeling problems into tractable computational workflows.

- Built an automated and hardware-accelerated mRNA design workflow for protein-coding tasks with realistic thermodynamics. We sample 2500 candidate sequences for the SARS-CoV-2 spike protein in about 18 seconds under an expressive free-energy model; LinearDesign takes ~11 minutes for a single MFE design under a simpler model.
- Built automatic tools for compressing Markov processes, backed by rigorous approximation theory and fast algorithms.
- Introduced an entropy-regularized semidefinite programming framework, coupled with scalable randomized algorithms, to tackle hard combinatorial optimization problems.
- Developed adaptive discretizations for quantum chemistry in a discontinuous Galerkin framework supported by fast adaptive solvers. Related work develops automatic GPU-oriented solvers for PDEs with applications from electronic structure to fluid dynamics.

Additional Technical Highlights

- Diffusion-based neural sampling and stochastic interpolants for multimarginal generative modeling.
 - Contributed to scientific machine learning through work on implicit bias in deep networks, tensor-network generative models, and scalable Gaussian process methods.
 - Publications spanning math, chemistry, physics, and machine learning, including papers in ICML, ICLR, NeurIPS, *Nature Communications*, and *Communications on Pure and Applied Mathematics*.
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Positions

University of California, Berkeley — Assistant Professor, Department of Mathematics, 2022–present

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Lawrence Berkeley National Laboratory — Faculty Scientist, AMCR Division, 2023–present

Courant Institute, New York University — NSF Math. Sciences Postdoctoral Research Fellow, 2019–2022

Leadership, Open Source, and Communication

- Co-PI on interdisciplinary funded projects in quantum computing (Department of Energy) and Bayesian digital twins (National Science Foundation).
 - Built interdisciplinary collaborations across mathematics, chemistry, physics, and machine learning.
 - Contributed open-source scientific code and research software; selected projects and code links are available on my website.
 - Mentored students and postdocs who have been placed in positions including the Courant Instructorship and Alvarez Fellowship (premier postdoctoral placements in applied math) and at Thinking Machines Lab (AI research lab).
 - Wrote pedagogical material including a textbook on continuous optimization.
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Selected Publications

- Direct RNA sequence design under codon constraints using expressive tensor-based secondary structure models (with Mark Fornace and Christina Wuyan Wang), 2026
 - An approximation theory for Markov chain compression (with Mark Fornace), 2025
 - Column and row subset selection using nuclear scores: algorithms and theory for Nyström approximation, CUR decomposition, and graph Laplacian reduction (with Mark Fornace), anticipated *Foundations of Computational Mathematics*, 2024
 - Implicit bias of gradient descent for non-homogeneous deep networks (with Yuhang Cai, Kangjie Zhou, Jingfeng Wu, Song Mei, Peter L. Bartlett), ICML, 2025
 - Multimarginal generative modeling with stochastic interpolants (with Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden), ICLR, 2024
 - Fast randomized entropically regularized semidefinite programming, accepted in *SIAM Journal on Scientific Computing*
 - Non-Hertz-Millis scaling of the antiferromagnetic quantum critical metal via scalable Hybrid Monte Carlo (with Peter Lunts, Michael S. Albergo), *Nature Communications*, 2023
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Awards and recognition

- Sloan Research Fellowship, 2024
 - Hellman Fellowship, 2024
 - John Todd Award (*numerical analysis*), Oberwolfach Foundation, 2022
 - NSF Mathematical Sciences Postdoctoral Research Fellowship, 2019
 - NSF Graduate Research Fellowship, 2016
 - Kennedy Thesis Prize (*top undergraduate thesis in natural sciences*), Stanford University, 2015
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Education

University of California, Berkeley — PhD, Applied Mathematics, 2019

Stanford University — BS with Honors, Mathematics, 2015